



without generating strains that compromise the electronic circuitry. So, engineers at UC Berkeley have developed a new, inexpensive technique for making wearable sensors that enable medical researchers to achieve their goals much faster. The new technique replaces photolithography—an expensive, multistep process for making computer chips. To achieve low-strain flexibility, an ‘island-bridge’ houses rigid electronics and sensor components. The bridges link the islands to one another. Their spiral and zigzag shapes stretch like springs to accommodate large deformations.

Low-energy chip that protects personal data

Side-channel attacks aim to gather secret information by indirectly exploiting a system or its hardware. Current prevention methods are power-intensive and unfeasible for

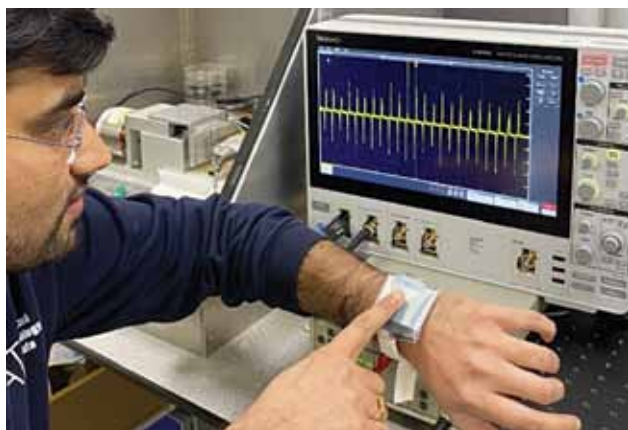


An application-specific integrated circuit chip that can be implemented on an IoT device to defend against power-based side-channel attacks (Credit: <https://news.mit.edu>)

low-power IoT devices like smartwatches. Now, engineers at MIT have built an integrated circuit chip that can defend against power side-channel attacks while using much less energy. The chip could be incorporated into a smartwatch, smartphone, or tablet to perform secure machine learning computations on sensor values. Rather than having a neural network operate on actual data, the chip is based on threshold computing in which the data are first split into unique, random components. The network operates on those random components individually, in a random order, before accumulating the final result.

Energy-harvesting wrist device made of discarded material

Scientists at the University of Surrey have unveiled a wrist device made from discarded paper wipes and plastic cups that runs on energy harvested by the wearer’s movements. The prototype device can transmit Morse code. The device is self-powered thanks to materials that become electrically



Wearable devices could soon be entirely made of recycled waste materials, and powered by human movement (Credit: www.surrey.ac.uk)

charged after they come into contact with one another. Known as triboelectric nanogenerators, they use static charge to harvest energy from movement through a process called electrostatic induction. The wearable device could be a future game-changer for the consumer, medical, and security sectors.

Printed transistors emulate neurons and synapses

In 2018 the research group at Linköping University developed complementary and printable organic electrochemical circuits, which made it possible to build printed complementary organic electrochemical transistors. The group had subsequently optimised the organic transistors, so that they



Artificial organic neurons (Credit: <https://liu.se/en>)

can be manufactured in printing presses on thin plastic foil. Now the group has used the printed transistors to emulate the neurons and synapses of the biological system. The hope is that artificial nerve cells can be used for sensitive human prostheses, implantable systems for relieving neurological diseases, and soft intelligent robotics.